Metrological Characterisation of a Low-Cost Ultrasonic Indoor Positioning System on a Mobile Robot and Correction of a Beacon Configuration Error

[TODO: working title]

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***Abstract*—[TODO: 150–200 words, write last. Problem: low-cost ultrasonic indoor positioning on a mobile robot, reference only in part of the area. What was done: Marvelmind on a KELO robot against OptiTrack in two campaigns at the same robot poses (rotations, triangles at two speeds, rests; out-and-back test). Main numbers: precision $1.6\,\mathrm{mm}$ (Allan, $1\,\mathrm{s}$) vs accuracy 3–7 cm at rest; about 11 cm RMS while moving. A wrong height of one beacon raised the p95 error up to 1.4 m; recomputing the recorded positions with the correct map, without the reference, made the first campaign indistinguishable from the second one, recorded with the correct configuration.]**

***Index Terms*—indoor positioning, ultrasonic positioning, motion capture, measurement uncertainty, mobile robots**

I. Introduction

[TODO: Low-cost indoor positioning systems (IPS) for mobile robots; their accuracy must be measured on the robot. Motion capture gives sub-millimetre reference positions, but only in a small volume.]

[TODO: Configuration of ultrasonic IPS (beacon positions and heights) is done by hand; a single wrong parameter is easy to miss and degrades the measurement. Can the data be saved afterwards?]

Contributions:

- [TODO: characterisation of Marvelmind on a mobile robot against OptiTrack: at rest (precision vs accuracy, repeated visits, heading), while moving, outside the reference area;]
- [TODO: effect of a wrong beacon height on the measurement;]
- [TODO: a method to recompute the recorded positions with the corrected map, without the reference, using the known height of the mobile beacon as a check, validated by a second campaign at the same robot poses with the correct configuration.]

II. Related Work

[TODO: Indoor positioning technologies and surveys [1], [2].] [TODO: Published evaluations of Marvelmind and other ultrasonic IPS: search IEEE Xplore and add.] [TODO: Motion capture as reference and its own accuracy [3].] [TODO: Trajectory error metrics (ATE) [4]; uncertainty evaluation [5]; Allan deviation [6].]

III. Experimental Setup

A. Marvelmind

[TODO: Four stationary beacons (B1, B2, B4, B5) at 2.40–2.60 m, Super-Beacon hardware [TODO: check], firmware 8.42, 3D navigation, update rate $5.8\,\mathrm{Hz}$, filters off, air temperature setting $23\,^{\circ}\mathrm{C}$. Hedgehog on the robot, read through the modem (binary protocol, device time stamp). Fig. 1(a).]

B. Reference: OptiTrack

[TODO: V120:Trio, $120\,\mathrm{Hz}$, Motive 3.5; covered area about $2.5\,\mathrm{m} \times 2.5\,\mathrm{m}$ in a corner of the beacon area; rigid body with 13 markers; static noise below $0.1\,\mathrm{mm}$, mean marker error about $1\,\mathrm{mm}$.]

C. Robot and mounting

[TODO: KELO driven with a joystick. Two hedgehog mountings; lever arm with respect to the rigid body pivot $(-0.144, -0.141)\,\mathrm{m}$ and $(0.214, 0.241)\,\mathrm{m}$, estimated from the data ($52\,\mathrm{cm}$ apart; tape: about $45\,\mathrm{cm}$). Rigid body: two groups of 5 markers on the mast and 3 markers $18.5\,\mathrm{cm}$ lower, at the height of the hedgehog ($0.69\,\mathrm{m}$ from the floor, $0.70\,\mathrm{m}$ reported by Marvelmind). Fig. 1(b).]

D. Logging and synchronisation

[TODO: Both systems on the same PC (`record.py`); Marvelmind device clock drift corrected; residual delay $0.09\,\mathrm{s}$.]

E. Test campaigns

[TODO: Two campaigns at the same robot poses (Table I). Campaign 1 is the main test programme; it was recorded with an error in the Marvelmind map (the height of beacon 1, $2.6\,\mathrm{m}$, was not applied and the map used $0\,\mathrm{m}$), found afterwards from the height reported for the mobile beacon. Campaign 2 repeats the rotations, the triangle and the rest with the correct map:

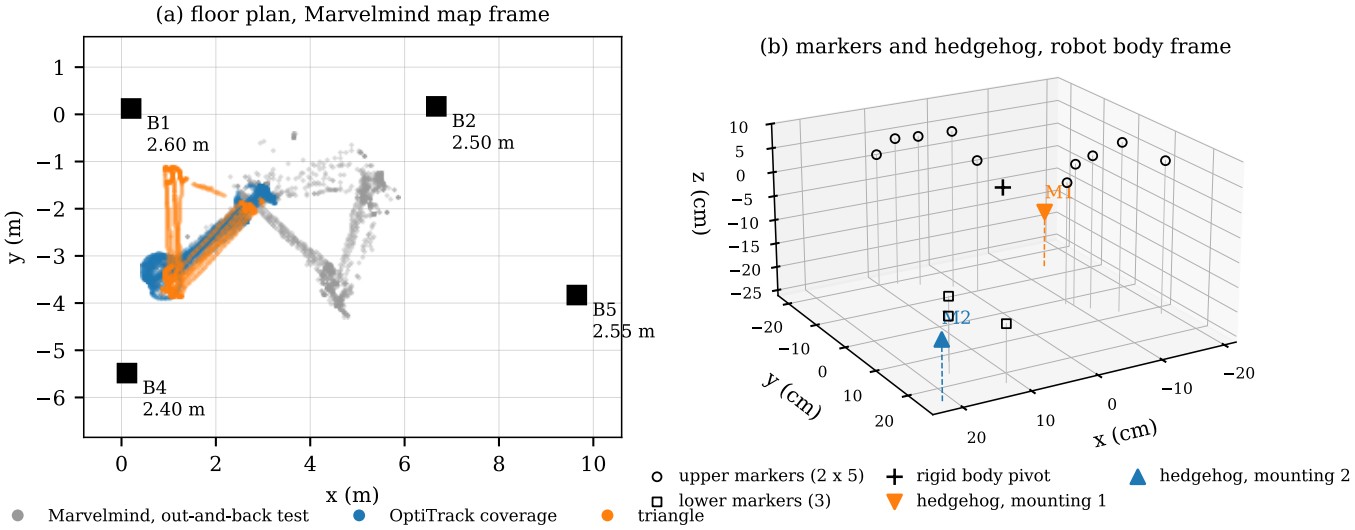


Fig. 1. Experimental setup. (a) Floor plan with the stationary beacons (height above the floor), the area covered by OptiTrack, the triangle path and the out-and-back path. (b) OptiTrack markers on the robot, rigid body pivot and position of the Marvelmind hedgehog in the two mountings (robot body frame). [TODO: add the photo of the robot]

TABLE I
TEST CAMPAIGNS. THE ROBOT POSES (TRIANGLE VERTICES AND HEADINGS) ARE THE SAME IN BOTH.

Test	Campaign 1	Campaign 2
Rotation in place, 4 headings	2 runs	2 runs
Triangle, 3 vertices	fast + slow	1 run
Rest	18 min	10 min
Out-and-back, 10 laps	yes	–
Marvelmind map	B1 height wrong	correct
Use	characterisation	alignment, validation

TABLE II
MARVELMIND ERROR (CM): CAMPAIGN 1 AS RECORDED (C1 REC.), CAMPAIGN 1 CORRECTED (C1 CORR.) AND CAMPAIGN 2 (C2). STOPS: MEDIAN / P95 OF THE PER-STOP OFFSET; MOVING: RMS / P95. LAST COLUMN: DIFFERENCE OF THE MEDIANS C1 CORR. − C2 WITH 95 % BOOTSTRAP INTERVAL.

Test	C1 rec.	C1 corr.	C2	C1 corr. − C2
Rotation, stops	16.6 / 85.8	4.0 / 12.1	7.4 / 13.5	−3.4 [−10.1, 2.2]
Triangle, stops	20.7 / 126.7	7.8 / 14.2	4.9 / 13.3	2.8 [−5.1, 5.4]
Triangle, moving	65.4 / 142.2	11.1 / 21.2	10.8 / 21.6	–
Rest	35.9	2.6	7.4	–

it is used to fit the alignment and as the reference for the correction of campaign 1.]

IV. METHOD

A. Correction of the beacon height error

[TODO: Each Marvelmind position comes from three beacons. From the recorded position and the wrong map, the distances to each triple are rebuilt; the position is recomputed with the corrected map for every triple; the triple whose height matches the known hedgehog height is kept, with a continuity term between samples (dynamic programming). The reference is not used. Checks: the four distances of one sample shown by the Marvelmind software fit the corrected map within 4 mm; the corrected campaign 1 is compared with campaign 2 (Section V-A).]

B. Alignment and delay

[TODO: 2D rotation, translation and lever arm, robust least squares [7], fitted on the stops of campaign 2 only (lever arm of the first mounting on the campaign 1 rotations, with rotation and translation fixed); delay fitted on the campaign 2 triangle. Triangles, out-and-back and rests of campaign 1 are test data.]

C. Data selection and metrics

[TODO: Stops: OptiTrack speed below 2 cm/s and rotation below 3 °/s for at least 10 s; frames with marker error above 3 mm discarded. Metrics: 2D error, median, RMS, p95, CDF, per-stop offset and jitter, Allan deviation [6].]

D. Uncertainty budget

[TODO: OptiTrack, lever arm, delay residual × speed, temperature (speed of sound, about 0.17 %/°C) [5].]

V. RESULTS

A. Effect of the configuration error and its correction

[TODO: Table II, Fig. 2: the configuration error multiplies the errors by 3–10; after the correction campaign 1 matches campaign 2 (differences of the median offsets with 95 % bootstrap intervals that include zero; moving errors equal within 0.4 cm). 81–100 % of the samples remain valid; one stop out of 23 failed the correction (167 cm).]

B. At rest

[TODO: Precision vs accuracy (Fig. 4): same noise in both campaigns, jitter 4 mm, Allan deviation 1.6 mm at 1 s and

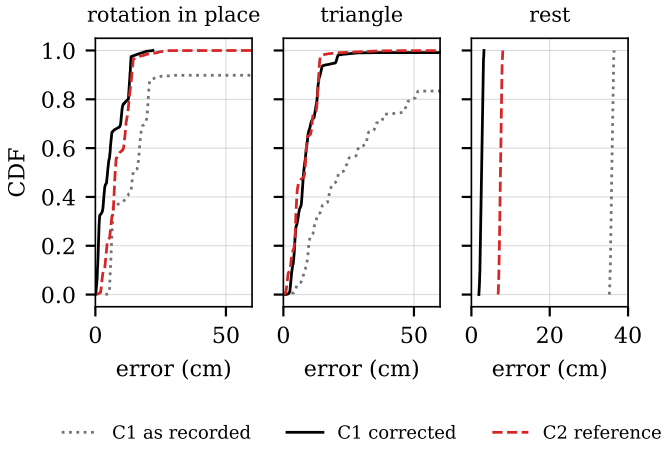


Fig. 2. Error distribution per test type: campaign 1 as recorded (dotted), campaign 1 after the correction (solid) and campaign 2 with the correct configuration (dashed).

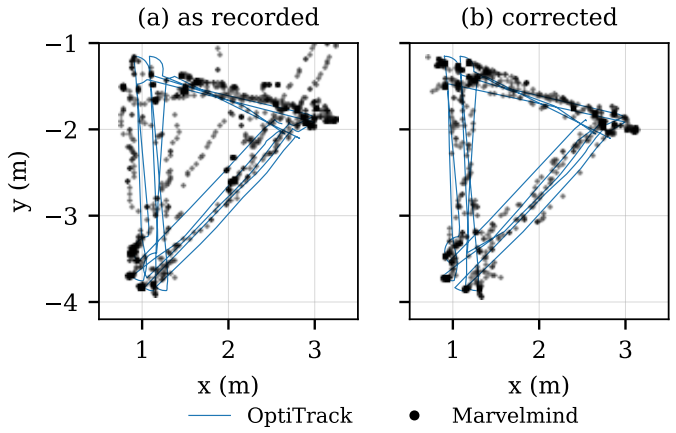


Fig. 3. Campaign 1, slow triangle: Marvelmind positions (dots) against the OptiTrack reference (line), (a) as recorded with the wrong height of beacon 1, (b) after the correction. [TODO: optional]

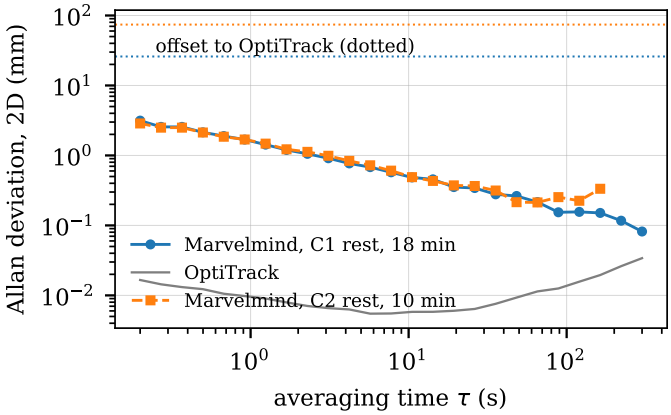


Fig. 4. Allan deviation of the Marvelmind position at rest in the two campaigns, and of OptiTrack. The dotted lines give the offset of each recording from the reference.

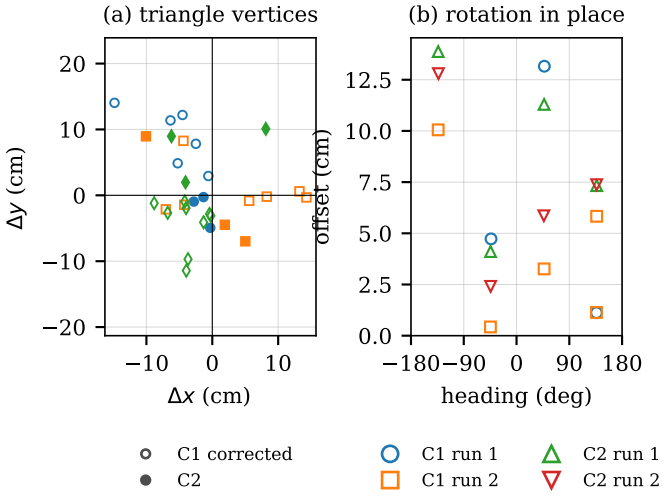


Fig. 5. Offset at rest. (a) Offset at each stop on the three triangle vertices, campaign 1 corrected (open) and campaign 2 (filled). (b) Offset against the robot heading for the rotation in place, two runs per campaign.

0.2 mm at 60 s; offset 2.6 and 7.4 cm. Repeated visits (Fig. 5a): offset per vertex 1–10 cm, spread between visits 3–9 cm, jitter within a stop 1.4 mm. Heading (Fig. 5b): the same pattern in both campaigns (−44°: 0.4–4.7 cm; −134°: 10–14 cm), so it belongs to the system.]

C. While moving

[TODO: RMS about 11 cm and p95 about 21 cm in both campaigns (Table II); no clear dependence on the speed up to 0.6 m/s (Fig. 6); delay 0.09 s. Temporal filters gained at most 1 cm.]

D. Outside the OptiTrack area

[TODO: Campaign 1 out-and-back: error before leaving 5.3 cm, after re-entry 4.3 cm (median, 10 re-entries); spread at the outside points 6–10 cm, an upper bound (Fig. 7).]

VI. Discussion and Conclusion

[TODO: Lessons for users: check that the mobile beacon reports its real height before recording; log the raw distances so that configuration errors can be corrected exactly. Limits: one room, 2D only, joystick paths, small OptiTrack area in a corner of the beacon area, temperature not measured.]

References

[1] R. Mautz, "Indoor positioning technologies," Habilitation thesis, ETH Zurich, 2012.

[2] F. Zafari, A. Gkelias, and K. K. Leung, "A survey of indoor localization systems and technologies," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2568–2599, 2019.

[3] P. Merriaux, Y. Dupuis, R. Boutteau, P. Vasseur, and X. Savatier, "A study of Vicon system positioning performance," *Sensors*, vol. 17, no. 7, p. 1591, 2017.

[4] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, "A benchmark for the evaluation of RGB-D SLAM systems," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 573–580.

[5] JCGM, "Evaluation of measurement data – guide to the expression of uncertainty in measurement," JCGM 100:2008, 2008.

[6] D. W. Allan, "Statistics of atomic frequency standards," *Proc. IEEE*, vol. 54, no. 2, pp. 221–230, 1966.

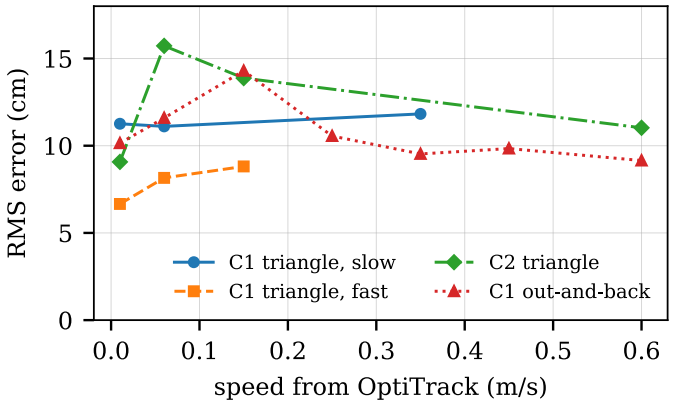


Fig. 6. RMS error of Marvelmind as a function of the speed measured by OptiTrack. [TODO: optional]

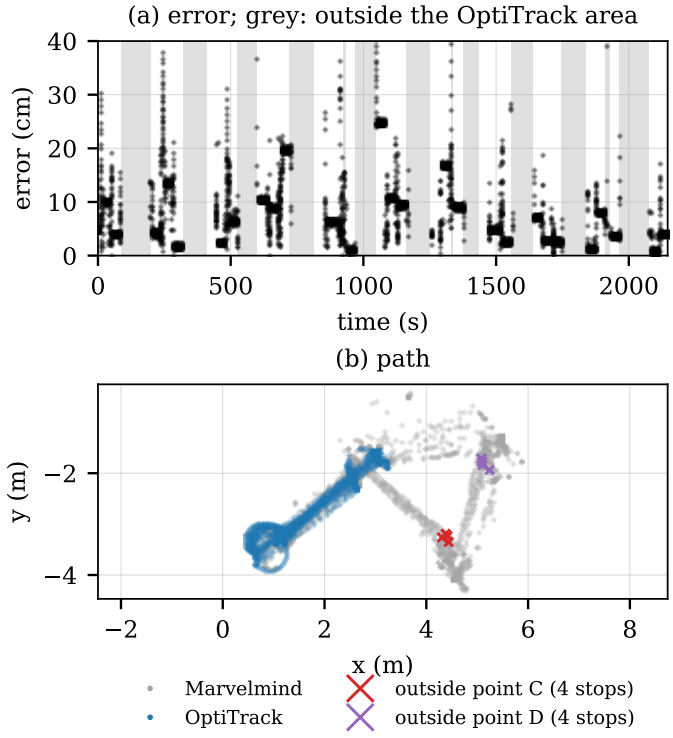


Fig. 7. Out-and-back test. (a) Marvelmind error over time; grey: outside the OptiTrack area. (b) Paths and Marvelmind positions at the outside stops.

[7] S. Umeyama, "Least-squares estimation of transformation parameters between two point patterns," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 13, no. 4, pp. 376–380, 1991.